AUDIT_SRC table info

There are four tables with an “oracle_” prefix. These table all live in the data lakehouse where a nightly data pipeline harvests the data from each of the 25 Oracle databases, which make up Cornell’s legacy Data Warehouse. Each of these four tables has a columns named “database_name”, which indicates which database, the data was harvested from. Note you will need to use the database_name column as an additional join, when using these tables, in order to gain uniqueness.

Three of the Oracle_ tables contain security access mapping data:

Oracle_users: This table contains a record for each defined database account in each database. Note that some of these accounts are delivered with Oracle and have never been used. There are some accounts which may not have access to any roles, so they will not show up in the audit table, or role_privs table. This is one of the exceptions, the CISO was interested in identifying. You will need this table in order to identify these accounts (usernames) which do not have access to any roles.

Note the in Oracle databases, a database account name is called “username”.

Oracle_role_privs: This table contains a record for each database_name, user_name (account_name), granted_role (database role) mapping in the data warehouse environment. *Note, for the class we only have data for two databases.*

Oracle_roles_to_tables: This table contains a record for each database_name, database_role (containing grants), owner (owner of a table or view), table_name (table or view, role has access to), privilege (SELECT, INSERT, UPDATE, DELETE).

Note that you will need to join the Oracle_role_privs and oracle_roles_to_tables, in order to get the mappings from database accounts (usernames) -> to roles –> to individual owner.table_names. You could also have an outer join from oracle_users to oracle_role_privs to identify the usernames with no role access.

The final Oracle table contains database audit data, which contains the needed usage data, to meet our sponsor’s requirements.

AUDIT_SRC table info

Oracle_table_audit: This table contains one record per database, per username (database account), query_date, action_name (SELECT, INSERT,UPDATE), object_name (table_name, view_name), query_count (total queries for that query_date). *Note if a user runs a query against 5 tables, it would result in 5 records in this table (each with a query count of 1).*

The remaining tables provide needed supplimental data.

Datamart_map: This table contains the mappings of database_name & owner (database schema) to datamart_name, datamart_abrv, environment (dev, test, prod), and datamart_subject.

Note that data marts are a logical structure, not known by the Oracle database, so this table will be needed in order to gain the full hierarchy of database-> datamart -> role -> table_name. This table also contains the environment (dev, test, prod) which was requested by the sponsors, and nice descriptive attributes for a dimension.

Datamart_roles: This table contains the Oracle database_role names (not user friendly) to role_descriptions (user friendly).

Note this table only contains the role mappings for the two databases we have security and audit data for class project. There are more roles in these databases, but they are not related to the datamarts. You will need an outer join from the oracle_role_privs to this table, so you don't lose some of the granted roles.

Cu_person: This is a subset of the records and columns, from Cornell's conformed person dimension table. This table can provide the needed owner name (first_name & last_name), unit_name, department_name, and employee status (active = A, I, R, N). A= Active, I= was an active employee no longer active, R = retired, N = never an active employee). For personal accounts, the cu_person.netid = oracle_users.username. For generic (or application) database accounts, you will see some of them in cu_person, but they do not contain the owner, or the needed unit and department.

AUDIT_SRC table info

Appid_owners: Per our meeting with our project sponsors, we did not have the account owner information for application accounts. I worked with the sponsors to locate this data, it can now be found in table appid_owners. The account_name is the non-netid database account, which could be joined to the Oracle_ tables. The owner_netid is the netid of the account owner. This may be joined to the cu_person.netid to get the first & last name, plus the org and department for the account owner. One thing to note, this table contains MOST of the application account owners, there will be some database accounts, which do not have a mapping. You will need to either exclude these accounts (and document this) or use an outter join and come up with default values for these ownerless accounts.